

STUDY QUESTIONS 2

1. An electronics assembly plant sends the produced parts to a test station before shipment after they leave the assembly line. At this station, it is checked whether the products are working or not. The production manager has noticed that there have recently been occasional buildups in the test department and wants to see whether the current capacity of this station is sufficient.

Parts arrive from the assembly line to the test station at an average rate of 30 parts per hour. However, the testing process is not perfect. Of the tested parts, 20% fail the first inspection and, after the necessary corrections are made, return to the queue of the same test station to be tested again. In other words, some parts cannot leave the system on the first pass and place an additional load on the test station.

There are 2 test machines operating at the same time in the test station. Each machine can test an average of 25 parts per hour.

- a) Calculate the effective arrival rate at the testing station (including initial arrivals + rework).
 - b) Calculate the utilization of the testing station.
2. A fast-growing logistics company is making capacity planning for a new distribution warehouse. The management team wants to evaluate three different effects together while calculating the warehouse's total annual cost. First, as long as the warehouse stays open, there is a fixed operating cost of 10,000 units each year. In addition to this, as the amount of capacity processed in the warehouse increases, a variable cost of 5 units occurs for each unit of capacity. However, the managers think that these two cost items alone are not enough, because as the warehouse usage level increases and the system gets closer to the maximum capacity limit of 200 units, serious congestion occurs in the operation. It is accepted that this additional cost, which appears because of reasons such as increased forklift traffic, workers waiting for each other, and longer product placement and order picking times, rises as the used capacity increases and grows much more sharply as the empty space remaining to the capacity limit becomes smaller. The company is planning its operation for the coming period at a capacity level of 100 units.

- a) Calculate the total annual cost ($C(q)$).

3. GreenTech Refurbishment Center specializes in restoring used laptops through a sequential two-stage workflow. Upon arrival, laptops first enter Station 1 (Diagnostics), where they are cleaned and undergo initial testing. After passing this stage, they move to Station 2 (Repair & QC) for necessary repairs and a final Quality Control check. A critical feature of this process is the internal rework loop: any laptop that fails the QC check is not sent back to the beginning but is immediately re-processed at Station 2, occupying the station's capacity again.

Under normal operating conditions, used laptops arrive at the facility at a rate of 30 laptops per hour. Currently, Station 1 is staffed by 3 technicians, each capable of processing an average of 12 units per hour. Station 2 is staffed by 2 senior engineers, each with a processing rate of 20 units per hour. Historical data indicates that 10% of the laptops fail the final quality control at Station 2 and require immediate rework.

- a) Calculate the Utilization Rate (ρ_1) for Station 1.
 - b) Calculate the Utilization Rate (ρ_2) for Station 2, accounting for the rework loop.

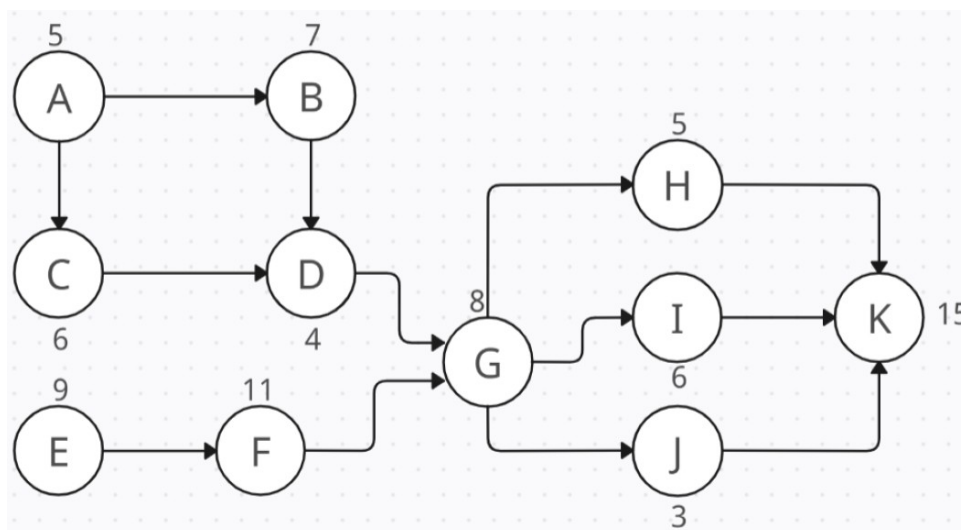
The facility is now facing a "Back-to-School" rush, causing significant operational strain. The arrival rate has surged to 38 laptops per hour. Furthermore, due to the increased pressure to work faster, the rework rate at Station 2 has doubled to 20%. Station 1 technicians maintain their normal processing rate of 12 units per hour each. Station 2 engineers also maintain their rate of 20 units per hour each. To mitigate these issues, the Operations Manager has secured a budget to hire one additional employee and must choose between two strategic options:

- **Option 1:** Hire an additional technician for Station 1 (bringing the total to 4 technicians, each processing 12 units/hour)

- **Option 2:** Hire an additional senior engineer for Station 2 (bringing the total to 3 engineers, each processing 20 units/hour)
 - c) For Option 1, calculate the utilization rates for both stations and determine the maximum system throughput (the maximum rate at which laptops can exit the system).
 - d) For Option 2, calculate the utilization rates for both stations and determine the maximum system throughput.
 - e) Based on your calculations, which option yields a higher production rate? Explain your reasoning and identify which station acts as the system bottleneck in each scenario.
4. **Baking a Cake:** The work elements, precedence requirements, and time requirements to bake a cake are shown here.
- a) Construct a precedence diagram of the process and label task times.
 - b) Set up an assembly line capable of producing 2,000 cakes per 40-hour week.
 - c) Calculate the efficiency and idle time of the line.

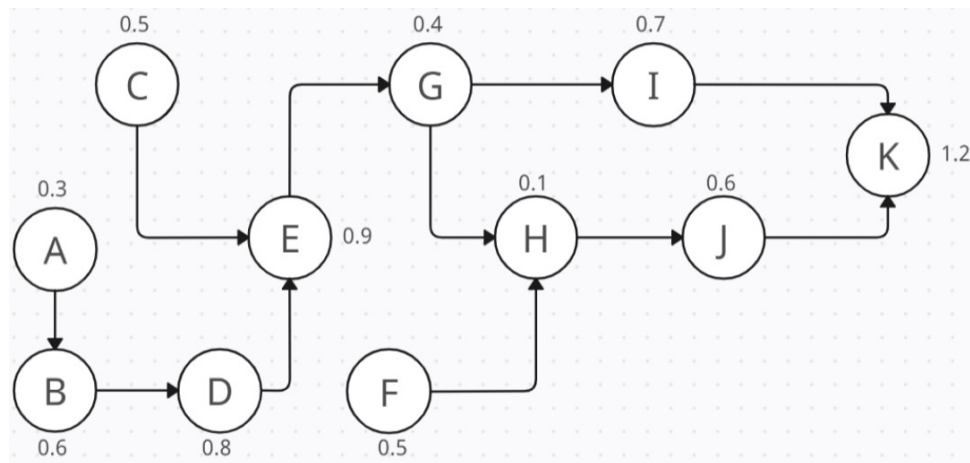
Element	Description	Precedence	Time(minutes)
A	Measure and prepare ingredients	-	0.30
B	Preheat the oven to 175°C	-	0.10
C	Mix dry ingredients (flour, sugar, baking powder)	A	0.40
D	Mix wet ingredients (eggs, milk, butter, vanilla extract)	A	0.40
E	Combine wet and dry ingredients to form the batter	B, C, D	0.50
F	Grease and flour the cake pan	-	0.15
G	Pour the batter into the cake pan	E, F	0.20
H	Place the cake pan in the oven and bake	B, G	0.50
I	Let the cake cool on a wire rack	H	0.30

5. **Automotive Factory Balancing:** The precedence diagram and test times (in minutes) for assembling ATD Automotive's automobile are shown below. Set up an assembly line to produce 130 automobiles in a 50-hour week. Balance this assembly line by:
- a) Longest operation time
 - b) Shortest operation time
 - c) Greatest number of following tasks



6. **Convert to a table:** The precedence diagram and test times (in hours) for assembling the Sessan-tasette 67 Corsa Yacht are shown below.

a) Convert the information from the diagram into the table provided below.



Element	Precedence	Time(hours)
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		
K		

7. **Work and Time Study:** At SwiftTech Manufacturing, a skilled technician assembles high-performance gaming keyboards. The process involves attaching mechanical switches to the circuit board, securing keycaps, installing stabilizers, and assembling the outer casing (total of 4 elements). A time analyst at the company has conducted a study of this process over 30 cycles, recording elemental times and assigning performance ratings to each step.

Determine the standard time for assembling a keyboard, considering an allowance factor of 9%.

Element	$\sum$ hours	Performance Rating
1	9.54	1.15
2	6.87	0.95
3	16.45	1.10
4	10.96	1.00

8. **Time Study of Valentine's Day:** FlowerBasket is a small company which prepares bundles of different types of flowers such as lilac, dandelion, rose, and so on, by a staff of workers, who are paid on a bundle-rate basis. A time study job has been conducted by an engineering consulting firm using a subject of bundles of Valentine's Day. The observations from the time study for 8 cycles of the four-element job and performance rating for each element are given next. Using an allowance factor of 12 percent, compute the standard time for this job.

Element	Elemental Times (minutes)								Performance Rating
	1	2	3	4	5	6	7	8	
1	0.09	0.16	0.13	0.12	0.85	0.08	0.15	0.17	1.09
2	0.05	0.15	0.14	0.08	0.09	0.07	0.78	0.11	1.06
3	0.12	0.07	0.09	0.06	0.05	0.10	0.06	0.13	0.90
4	0.06	0.07	0.54	0.11	0.08	0.08	0.07	0.07	1.10

9. **Work and Time Study 2:** A paper factory specializes in producing various types of paper products, including packaging paper, office sheets, and specialty papers. Given the following data, compute the production standard (pieces per hour), the standard labor cost per 100 pieces, and the standard hours per 100 pieces. The allowances are 8% for repetitive task fatigue, 4% for extreme noise, and 6% for personal needs. The operator is paid \$10 per hour. Performance rating = 95%.

		Observation Results (minutes)				
		1	2	3	4	5
Element	1	0.036	0.190	0.147	0.121	0.156
	2	0.033	0.033	0.014	0.046	0.256
	3	0.142	0.194	0.006	0.178	0.167
	4	0.062	0.088	0.060	0.056	0.075

10. **Productivity:** You are working with ABIR, Inc., on productivity measurement. They give you the data below. For those data:

- Calculate partial productivity ratios for this year and the base year.
- Calculate productivity ratios for this year.
- What can you say about the company's productivity?

	Base Year	This Year
Output	50	70
Capital Investment	12,000	19,000
Overhead Cost	6,500	8,000
Labor Hours	265	310
Machine Hours	240	290